

Crashfree India

Join us in building India's Apex body in safer mobility

Thank you for applying to Crashfree India Internship Program (August 2025). As we dive into the **third round of assessment we seek for clarity, structure and relevance to the submissions made to the respective vertical on ensuring the focus on road-safety**. This submission round will serve as a testament to the applicant's significance for the aspired role based on relevant skillsets.

All submissions will remain the intellectual property of the applicant. Crashfree India (CFI) may use the submitted work only with the applicant's prior approval.

This is your chance to demonstrate creative thinking and a structured approach. A well-crafted, authentic submission not only reflects your capabilities but can also open meaningful opportunities ahead.

Location: **In-office (Gurgaon)** | Duration: **2 months, 3 months, 6 months** Start Date: **11/08/2025** | Stipend: **INR 7.5k - INR 12k** (*Basis Interview Performance*)
Perks: **LOR, mentorship, and a chance to join the CFI core team**

Profile: Founder's Office Intern

Assignment Details: You are given a dashcam video ([Link](#)) recorded from a moving vehicle on Indian roads. Your task is to build an ML/AI model pipeline that automatically analyzes the video and produces a structured dashboard with the following outputs:

Task 1 — Traffic Violation Detection

Detect and count the following five traffic violations as defined under the Motor Vehicles Act, 1988 and Central Motor Vehicles Rules, 1989:

1. Helmet-less riding — Two-wheeler rider or pillion without a helmet
2. Wrong-side driving — Vehicle moving against the flow of traffic on its lane
3. Signal jumping — Vehicle crossing a stop line while the signal is red
4. Mobile phone use while driving — Driver visibly operating a handheld device
5. Triple riding — Three or more persons on a two-wheeler

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For each violation, report the following:

- Total count across the entire video
- Timestamp of each detected instance (HH:MM:SS)
- Frame-level bounding box annotation for the top three confidence detections per violation type

Task 2 — Junction & Road Event Detection

Detect and count all junctions and road intersections the dashcam vehicle passes through. Classify each into one of the following types:

1. T-Junction — Three-arm intersection (most common in Indian towns)
2. 4-Way / X-Junction — Four-arm crossroad intersection
3. Y-Junction — Diverging or merging fork at an acute angle
4. Roundabout / Traffic Circle — Circular intersection with a central island
5. Flyover / Underpass Entry-Exit — Grade-separated crossing point

Report the following:

- Total junction count
- Type-wise breakdown
- Timestamps of each detected junction crossing

Task 3 — Vehicle Detection & Classification

Detect all vehicles visible in the video and classify them into the following categories per Indian vehicle classification standards:

1. Two-Wheeler (2W) — Motorcycles, scooters, mopeds
2. Light Motor Vehicle (LMV) — Cars, jeeps, taxis, auto-rickshaws
3. Heavy Motor Vehicle (HMV) — Trucks, buses, tankers, trailers
4. Others — Cycles, e-rickshaws, construction vehicles, unclassified

Report the following:

- Total unique vehicle count, de-duplicated across frames using object tracking
- Category-wise count and percentage share
- Instantaneous vehicle density over time (vehicles per frame)

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Task 4 — Analytical Dashboard

Build a dashboard (any frontend framework) that displays all of the above. The dashboard must include:

- Summary metric cards showing total violations, total junctions, and total vehicles
- A violations breakdown chart by violation type (bar or donut)
- A vehicle category distribution chart
- A timeline and event log showing all detected events with timestamps
- A junction log with type classification
- The ability to seek to a video timestamp by clicking any event in the log

Deliverables

1. Source code hosted on a public GitHub repository
2. Model weights and inference pipeline, runnable locally or via a hosted demo
3. Sample output — run on the provided test video, exported as JSON and rendered as a live dashboard
4. README with setup instructions, model choices, architecture overview, and known limitations
5. Brief report (maximum two pages) covering your approach, dataset used for training or fine-tuning, and model metrics (mAP, FPS, accuracy per class)

Please note that all submissions will be evaluated against our in-house model, which has been trained to detect the same set of violations, junctions, and vehicle categories outlined in this assignment. Your model's outputs — including counts, classifications, and timestamps — will be benchmarked directly against ours on the same test video. The closer your results are to our model's detections in terms of accuracy, consistency, and coverage, the higher your score will be.

Ensure that all assumptions, rationale, and key action points are clearly stated.

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Submission Guidelines

- **Submit all the given tasks in a separate document** all to be uploaded in a common drive link - ensure that the folder link has view access before submitting.
- Orient your work logically showcasing a **clear flow of ideas more than decorative language or lengthy writing**. Prioritise to use headers, bullet points, or sections to make your documents easy to follow.
- **Focus on depth, not length** as a sharp, well-argued 2-pager is better than a vague 6-page report. Make sure each section delivers value and relevance.
- You are encouraged to use AI tools (including ChatGPT) to support your research or editing — but **the thinking, analysis, and approach must be your own. AI is a supplement**, not a substitute for your original thinking.

Video Submission

Present your assignment overview In up to 2 minutes video: the idea and your approach. Keep it crisp, we will review the detailed document separately. We're not expecting you to cover every detail, but **show us the strength of your thinking, structure, and approach**. You can record this on your phone or laptop. Keep it simple and clear — **we're evaluating clarity of thought, not video production quality**.

How Your Assignment Will Be Evaluated

Each submission will be evaluated on the following parameters:

- Clarity of Concept & Alignment with brief
- Creativity and Originality in relation to CFI Approach
- Execution and Technical Proficiency
- Impact and Audience Relevance Created

Submit your detailed assignment (PDF/PPT) via this link: [Assignment Submission](#)

We're excited to learn more about your ideas and your story.
Let's build a crash-free India — together.